



NAME:
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INSTRUCTOR:

MATH 205 LINEAR ALGEBRA

MIDTERM PART A

[Total Marks: 50]

FALL 2022

Question 1: [10 Marks]

- (a) Suppose $AX = B$ is the matrix equation for a system of equations in m variables with A being an $m \times m$ coefficient matrix. Prove that if A is not singular, then this system has only one solution.

- (b) List the three possible elementary row operations and define an elementary matrix.



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Question 2: [10 Marks]

Let Y and Z be $n \times n$ matrices.

(a) Prove that if $I_n = YZ$ then $Z = Y^{-1}$.

(b) Prove that if $I_n = ZY$ then $Z = Y^{-1}$.



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Question 3: [10 Marks]

Consider a homogenous system in n variables for which the coefficient matrix is B .
Given a non-singular $n \times n$ matrix M , show that the $BX = 0$ has only the trivial solution
if and only if $(MB)X = 0$ only has a trivial solution.

Question 4: [10 Marks]

Show that the diagonal entries of a 3×3 skew-symmetric matrix are all zero.



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Question 5: [10 Marks]

Prove the following identity WITHOUT evaluating determinants.

$$\begin{vmatrix} a_1 & b_1 + ta_1 & c_1 + rb_1 + sa_1 \\ a_2 & b_2 + ta_2 & c_2 + rb_2 + sa_2 \\ a_3 & b_3 + ta_3 & c_3 + rb_3 + sa_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$



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MIDTERM PART B

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Question 6: [10 Marks]

Check whether the set of all triples of real numbers with the standard vector addition and with scalar multiplication defined by $k(x, y, z) = (k^2x, k^2y, k^2z)$ is a vector space or not. If not, identify all axioms that fail to hold.



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Question 7: [10 Marks]

Prove: If W is a set of one or more vectors from a vector space V , then W is a subspace of V if and only if the following conditions hold:

- (a) If $\underline{u}$ and $\underline{v}$ are vectors in W , then $\underline{u} + \underline{v} \in W$.
- (b) If k is any scalar and $\underline{u}$ is any vector in W , then $k\underline{u} \in W$.



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Question 8: [10 Marks]

(a) Indicate whether each of the following statements is always true or sometimes false.

Justify your answers:

- (i) If W is a set of one or more vectors from a vector space V , and if $k\underline{u} + \underline{v}$ is a vector in W for all vectors $\underline{u}$ and $\underline{v}$ in W and all $k \in \mathbb{R}$, then W is a subspace of V .

- (ii) If $\text{span}(S_1) = \text{span}(S_2)$ then $S_1 = S_2$.

- (b) In words, describe a set of matrices that spans M_{nn} .



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Question 9: [10 Marks]

Let $\underline{u}$ and $\underline{v}$ be vectors in the Euclidean space $\mathbb{R}^n$. Show the following:

(a) $A\underline{u} \cdot \underline{v} = \underline{u} \cdot A^T \underline{v}$.

(b) $\|\underline{u} + \underline{v}\| \leq \|\underline{u}\| + \|\underline{v}\|$.



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Question 10: [10 Marks]

Let V be the subspace of $F(-\infty, \infty)$ given by $V = \text{span}\{\sin x, \cos x\}$. Show that for any given value of θ , $f_1 = \cos(x + \theta)$ and $g_1 = \sin(x + \theta)$, we have that $f_1 \in V$ and $g_1 \in V$.



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Rough Work